Part 1 — Naive Bayes Classifier

Probability aur Bayes' Theorem par bana ek simple lekin bahut powerful classification algorithm.

1. Introduction — Naive Bayes Kya Hai?

Pichhle notes me humne Linear aur Logistic Regression padhi thi. Ab hum ek aur classification algorithm seekhenge — **Naive Bayes**. Classification matlab output ek **category** hota hai, jaise Yes/No, Spam/Not Spam, ya Play Tennis: Yes/No.

Naive Bayes ko samajhne ke liye pehle do cheezein clear honi chahiye — teacher ne bhi notes ke top par yahi do heading likhi hain:

- **Probability** — kisi event ke hone ke chances kitne hain.
- **Bayes' Theorem** — ek formula jo bataata hai ki "agar B ho chuka hai, to A ke hone ke chances kitne hain."

Naam me **"Bayes"** isliye kyunki algorithm Bayes' Theorem use karta hai, aur **"Naive"** (matlab bholi/simple) isliye kyunki ye ek simple assumption maan leta hai — ki saare input features ek-dusre se **independent** hain. Ye assumption aage detail me samjhenge.

> **Dhyaan dein:** Naive Bayes **Classification** algorithm hai. Andar se ye har class (jaise Yes aur No) ki probability nikalta hai, aur jis class ki probability sabse zyada hoti hai, wahi final prediction ban jaati hai.

2. Probability Basics — Independent aur Dependent Events

Probability ka basic formula: **P(event) = favourable outcomes / total outcomes**. Naive Bayes me do tarah ke events samajhna zaroori hai.

2.1 Independent Events

Independent event wo hota hai jiski probability pichhle event se **bilkul affect nahi hoti**. Teacher ka example: ek **dice roll** karna. Dice ke outcomes {1, 2, 3, 4, 5, 6} hain aur har number ka chance barabar hai:

$$P(1) = 1/6 \quad P(2) = 1/6 \quad P(3) = 1/6$$

Extra example (samajhne ke liye): agar dice ko do baar roll karein, to pehle roll me 6 aane se doosre roll me 6 aane ke chances nahi badalte. Isliye P(6 aur phir 6) = 1/6 × 1/6 = 1/36. Independent events me bas probabilities ko **multiply** kar dete hain: **P(A and B) = P(A) × P(B)**.

2.2 Dependent Events — Marbles Wala Example

Dependent event me doosre event ki probability **pehle event par depend** karti hai. Teacher ka sawaal: *"Ek box se pehle ek orange marble nikalne aur phir ek yellow marble nikalne ki probability kya hai?"* Box me kul **5 marbles** hain — **3 orange** aur **2 yellow**.

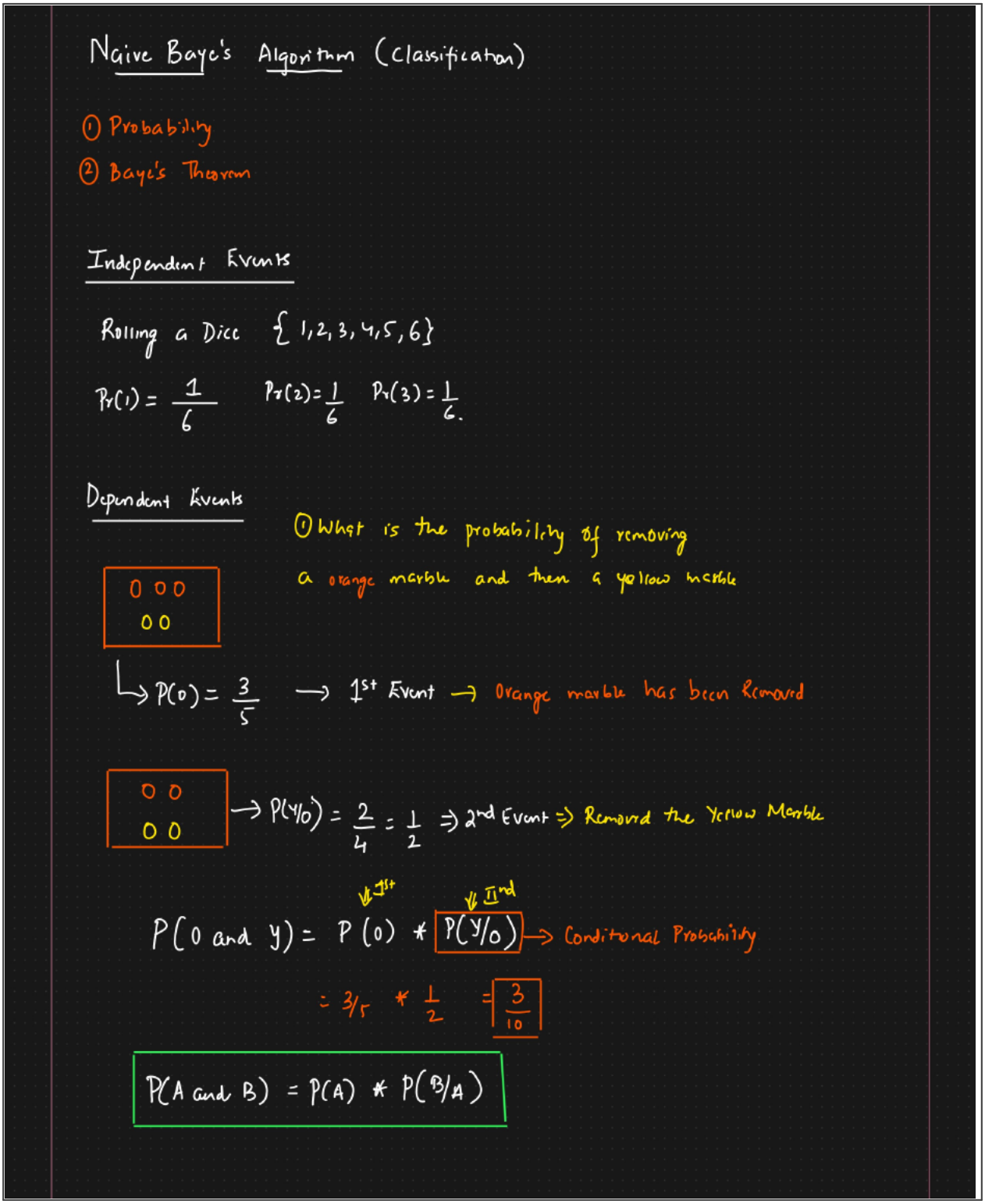
Step	Kya hua	Box me ab kya bacha	Probability
1st event	Orange marble nikala	3 orange, 2 yellow (kul 5)	P(O) = 3/5
2nd event	Ab yellow marble nikalna hai (orange wapas nahi daala)	2 orange, 2 yellow (kul 4)	P(Y | O) = 2/4 = 1/2
Dono saath	Dono probabilities multiply	—	3/5 × 1/2 = **3/10**

Yahan **P(Y | O)** ko **Conditional Probability** kehte hain. "|" ka matlab hai **"given that"** — yaani "orange nikal chuka hai, ye maan kar yellow ki probability." Isi se dependent events ka formula milta hai:

$$P(A \text{ and } B) = P(A) \times P(B \mid A)$$

Yaad rakhein: Independent events me P(B | A) = P(B) ho jaata hai (kyunki A se B par farq nahi padta), isliye formula P(A) × P(B) ban jaata hai. Dependent events me hume conditional probability use karni padti hai.

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Isme dikhaya gaya hai: Probability aur Bayes' Theorem ka agenda, Independent Events (dice), Dependent Events (marbles ka example) aur Conditional Probability ka formula P(A and B) = P(A) × P(B/A).

3. Bayes' Theorem — Formula Kaise Bana?

Ab hum upar ke formula se Bayes' Theorem step-by-step nikalte hain. Bas ek simple observation chahiye:

Step 1: "A aur B dono hona" aur "B aur A dono hona" ek hi baat hai, order se farq nahi padta:

$$P(A \text{ and } B) = P(B \text{ and } A)$$

Step 2: Dono taraf conditional probability wala formula laga do:

$$P(A) \times P(B \mid A) = P(B) \times P(A \mid B)$$

Step 3: Dono side ko P(B) se divide karo, to hume milta hai **Bayes' Theorem**:

$$P(A \mid B) = [\, P(A) \times P(B \mid A) \,] \,/\, P(B)$$

Formula ke har term ka matlab (teacher ke notes ke mutabik):

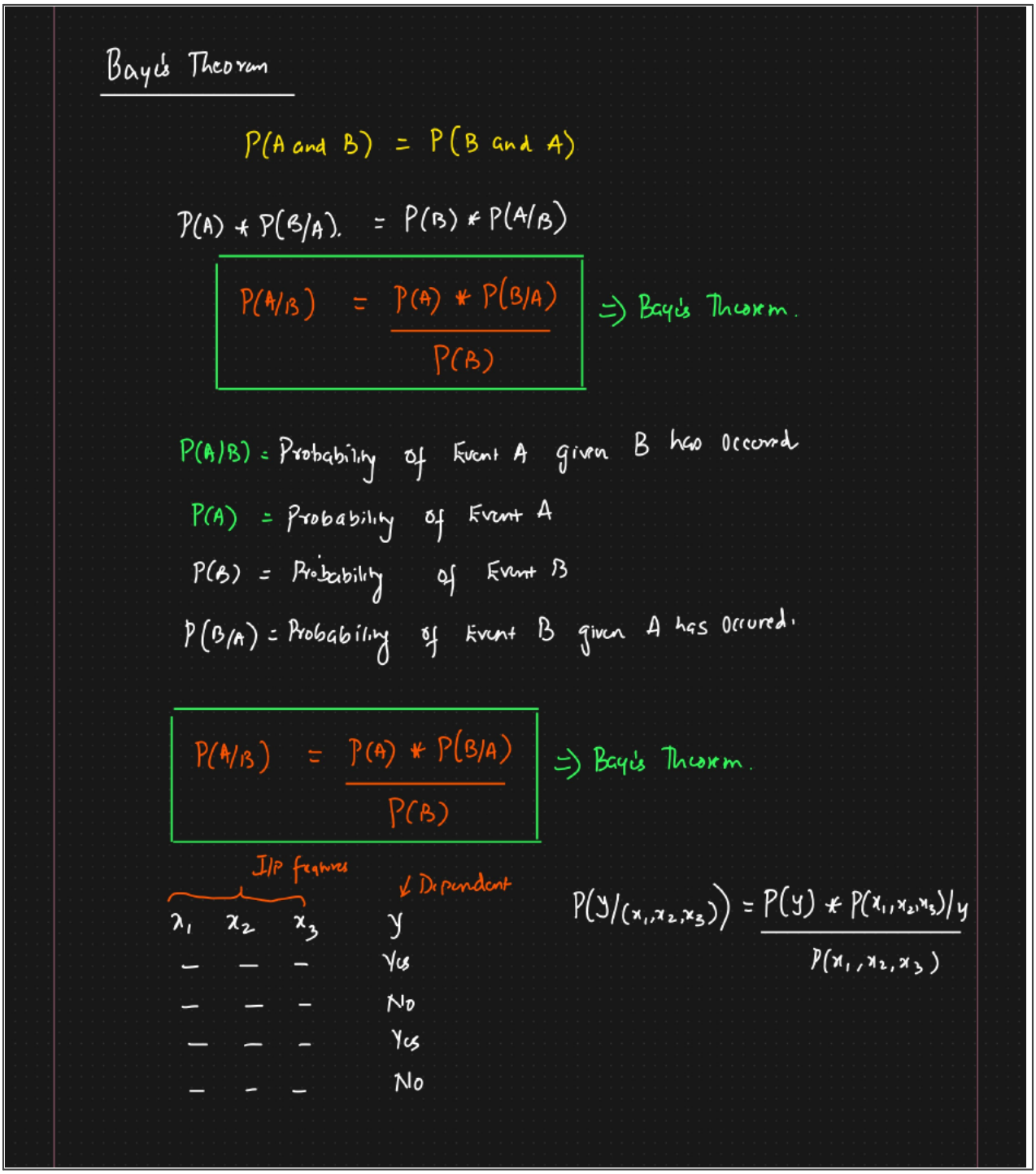
Term	Matlab	Common naam (extra)
P(A | B)	Event A ki probability, jab B ho chuka ho	Posterior
P(A)	Event A ki probability	Prior
P(B)	Event B ki probability	Evidence
P(B | A)	Event B ki probability, jab A ho chuka ho	Likelihood

3.1 Machine Learning Dataset par Bayes' Theorem

Ab maan lo humare paas ek dataset hai jisme input features $\mathbf{x_1}, \mathbf{x_2}, \mathbf{x_3}$ hain aur output (dependent variable) **y** hai (jaise Yes/No). Humein ye nikalna hai: "in features ke diye jaane par y = Yes hone ki probability kitni hai?" Bayes' Theorem me **A = y** aur **B = (x_1, x_2, x_3)** rakhne se:

$$P(\, y \mid x_1, x_2, x_3 \,) = [\, P(y) \times P(\, x_1, x_2, x_3 \mid y \,) \,] \,/\, P(\, x_1, x_2, x_3 \,)$$

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Isme dikhaya gaya hai: Bayes' Theorem ki derivation (P(A and B) = P(B and A)), final formula, har term ka meaning, aur ML dataset (x1, x2, x3 → y) par Bayes' Theorem ka application.

4. "Naive" Assumption — Formula Ko Simple Kaise Banaya?

Pichhle formula me ek badi dikkat hai: **$P(x_1, x_2, x_3 \mid y)$** — yaani teeno features ek saath ek particular combination me aane ki probability — data se nikalna bahut mushkil hai (features badhne par combinations bahut zyada ho jaate hain).

Isi problem ko solve karne ke liye Naive Bayes ek **"naive" assumption** leta hai: **class y diya ho, to saare features ek-dusre se independent hain.** Independent events me hum probabilities multiply kar sakte hain (section 2.1), to formula tootkar simple ban jaata hai:

$$P(y \mid x_1, x_2, x_3) = [\, P(y) \times P(x_1 \mid y) \times P(x_2 \mid y) \times P(x_3 \mid y) \,] \,/\, [\, P(x_1) \times P(x_2) \times P(x_3) \,]$$

4.1 Denominator Constant Hota Hai — Isko Hata Do

Ab maan lo humein ek **naye test data** (x_1, x_2, x_3) ke liye **Yes** aur **No** dono ki probability nikalni hai. Dono formulas me neeche ka denominator **bilkul same** hai, kyunki wo sirf features par depend karta hai, class par nahi. Isliye teacher ne usko cut karke "constant" likha hai — comparison ke liye uski zaroorat nahi:

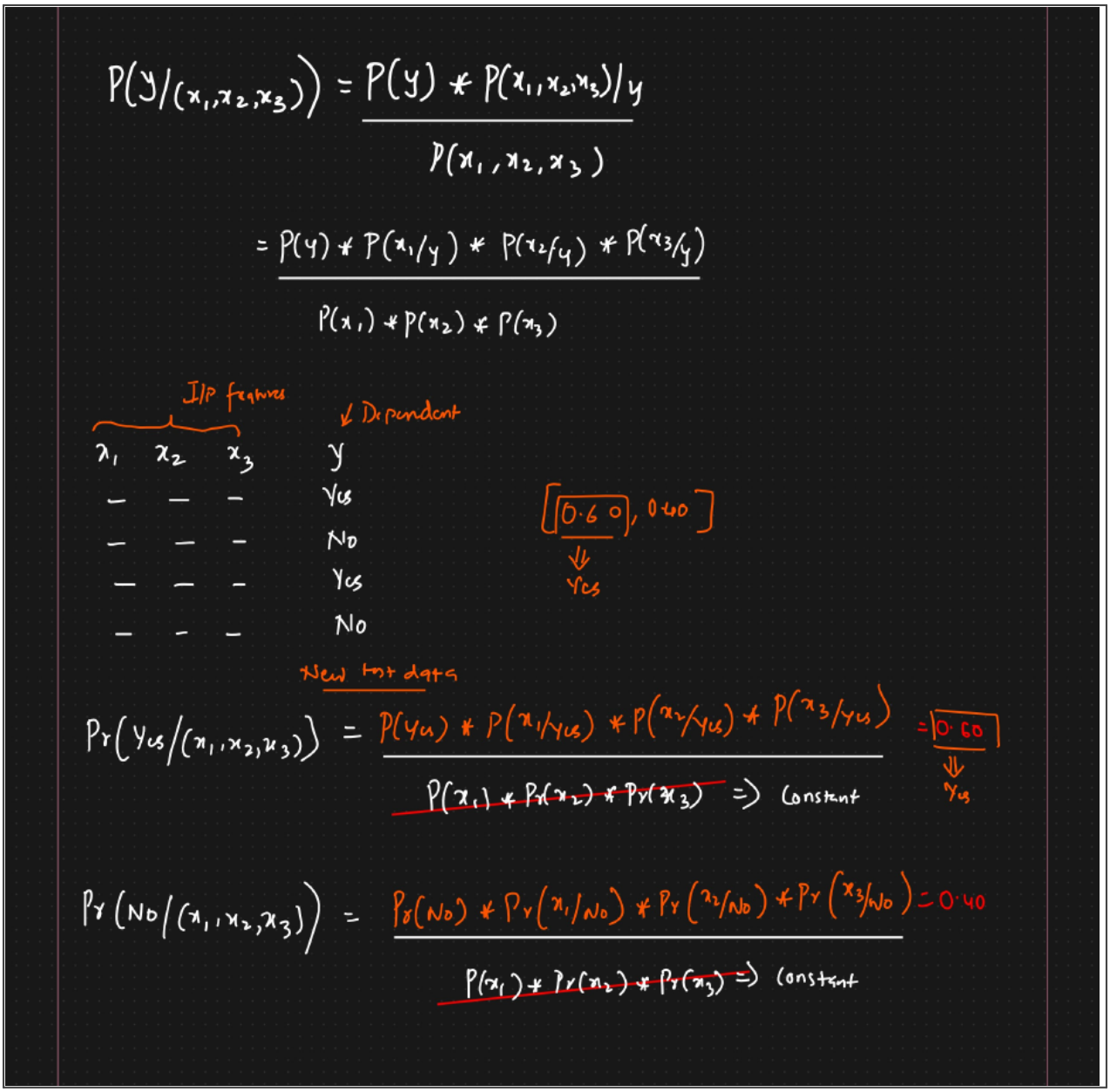
$$\text{Score(Yes)} = P(\text{Yes}) \times P(x_1 \mid \text{Yes}) \times P(x_2 \mid \text{Yes}) \times P(x_3 \mid \text{Yes})$$
$$\text{Score(No)} = P(\text{No}) \times P(x_1 \mid \text{No}) \times P(x_2 \mid \text{No}) \times P(x_3 \mid \text{No})$$

Yahan "Score" ka matlab hai denominator hata kar bacha hua numerator. Jis class ka score bada aaye, wahi prediction. Teacher ke example me aise do values aaye **[0.60, 0.40]** — 0.60 bada hai, isliye output = **Yes**.

4.2 Poora Algorithm — 5 Steps Me

- **Step 1:** Training data se har class ki **prior** probability nikalo — P(Yes), P(No).
- **Step 2:** Har feature ki har value ke liye **P(feature | class)** ki table banao.
- **Step 3:** Naye data ke liye, har class ke liye prior × saari conditional probabilities **multiply** karo.
- **Step 4:** (Optional) dono values ko unke sum se divide karke **normalize** karo, taaki percentage mil jaaye.
- **Step 5:** Jis class ki value sabse badi ho, wahi **predicted class** hai.

> **"Naive" kyun?** Real life me features aksar independent nahi hote (jaise Outlook aur Humidity thode connected ho sakte hain). Phir bhi independence maan lene se calculation bahut aasaan ho jaata hai aur practice me algorithm kaafi achha chalta hai — isi simplification ki wajah se naam "Naive" pada.



Isme dikhaya gaya hai: Bayes' Theorem ka expansion (P(y) × P(x1/y) × P(x2/y) × P(x3/y)), denominator ko constant maan kar cut karna, aur Yes/No dono ki probability compare karke [0.60, 0.40] → Yes predict karna.

5. Worked Example — Play Tennis Dataset

Ab ek poora example solve karte hain jo teacher ne notes me kiya hai. Dataset me 14 din ka record hai ki mausam ke hisaab se tennis khela gaya ya nahi:

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Sawaal: Naya test data hai **Outlook = Sunny, Temperature = Hot**. Kya person tennis khelega? (Teacher ne is example me sirf Outlook aur Temperature use kiye hain.)

Step 1 — Prior Probabilities

14 din me se **9 din Yes** aur **5 din No** hain:

$$P(Yes) = 9/14 \qquad P(No) = 5/14$$

Step 2 — Outlook Ki Conditional Probability Table

Har Outlook value ke liye count karo ki Yes me kitni baar aayi aur No me kitni baar. Phir Yes wale count ko 9 se aur No wale count ko 5 se divide karo:

Outlook	Yes	No	P(Outlook | Yes)	P(Outlook | No)
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rain	3	2	3/9	2/5

Step 3 — Temperature Ki Conditional Probability Table

Temperature	Yes	No	P(Temp | Yes)	P(Temp | No)
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
Cool	3	1	3/9	1/5

Teacher ke notes me small typo: Handwritten table me Mild ke liye P(Temp | No) = 4/5 aur Cool ke liye 3/5 likha hai. Sahi values **2/5** (Mild) aur **1/5** (Cool) hain, kyunki No wale 5 dino me Mild 2 baar aur Cool 1 baar aaya hai. Is example me sirf **Hot** use hua hai, isliye final answer par koi farq nahi padta. Exam me sahi values hi likhna.

Step 4 — "Yes" Ki Score Nikalo

Formula: P(Yes) × P(Sunny | Yes) × P(Hot | Yes). Denominator P(Sunny) × P(Hot) dono class me same hai, isliye cut kar diya:

$$\text{Score(Yes)} = P(\text{Yes}) \times P(\text{Sunny} \mid \text{Yes}) \times P(\text{Hot} \mid \text{Yes}) = 9/14 \times 2/9 \times 2/9 = 2/63 \approx 0.031$$

Step 5 — "No" Ki Score Nikalo

$$\text{Score(No)} = P(\text{No}) \times P(\text{Sunny} \mid \text{No}) \times P(\text{Hot} \mid \text{No}) = 5/14 \times 3/5 \times 2/5 = 3/35 \approx 0.085$$

(Exact value 0.0857 hai; teacher ne 0.085 likha hai.)

Step 6 — Normalize Karke Percentage Nikalo

Dono scores ko unke sum se divide karo, taaki dono milakar 100% ban jaayein:

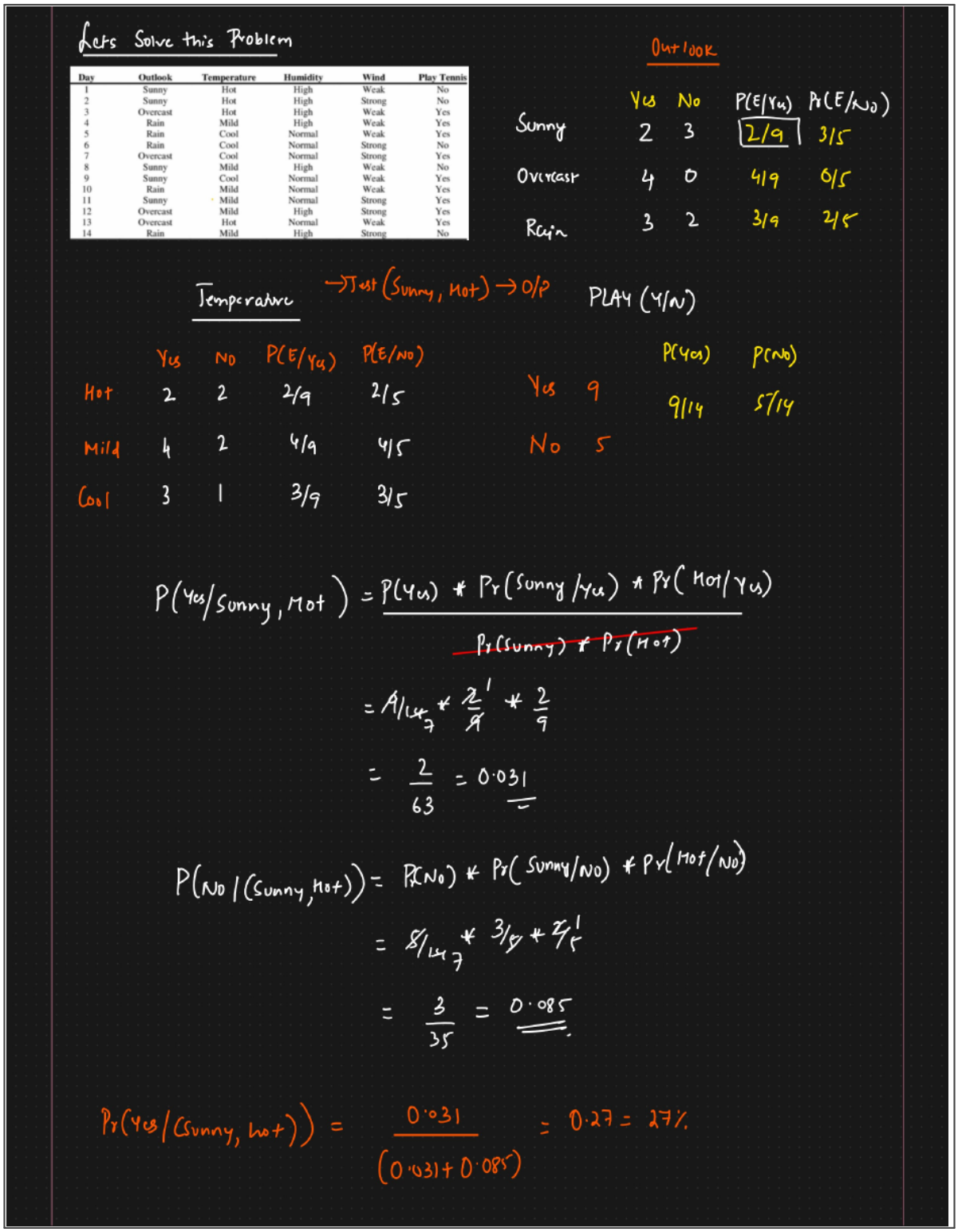
$$P(\text{Yes}) = 0.031 / (0.031 + 0.085) \approx 0.27 = 27\%$$
$$P(\text{No}) = 0.085 / (0.031 + 0.085) \approx 0.73 = 73\%$$

Step 7 — Final Prediction

Outlook	Temperature	Output
Sunny	Hot	73% No → person tennis nahi khelega (**0**) | 27% Yes

Yaad rakhein: Naive Bayes ka answer hamesha us class ka hota hai jiski (normalized) probability sabse zyada ho. Yahan 73% > 27%, isliye predicted class = **No**.

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Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

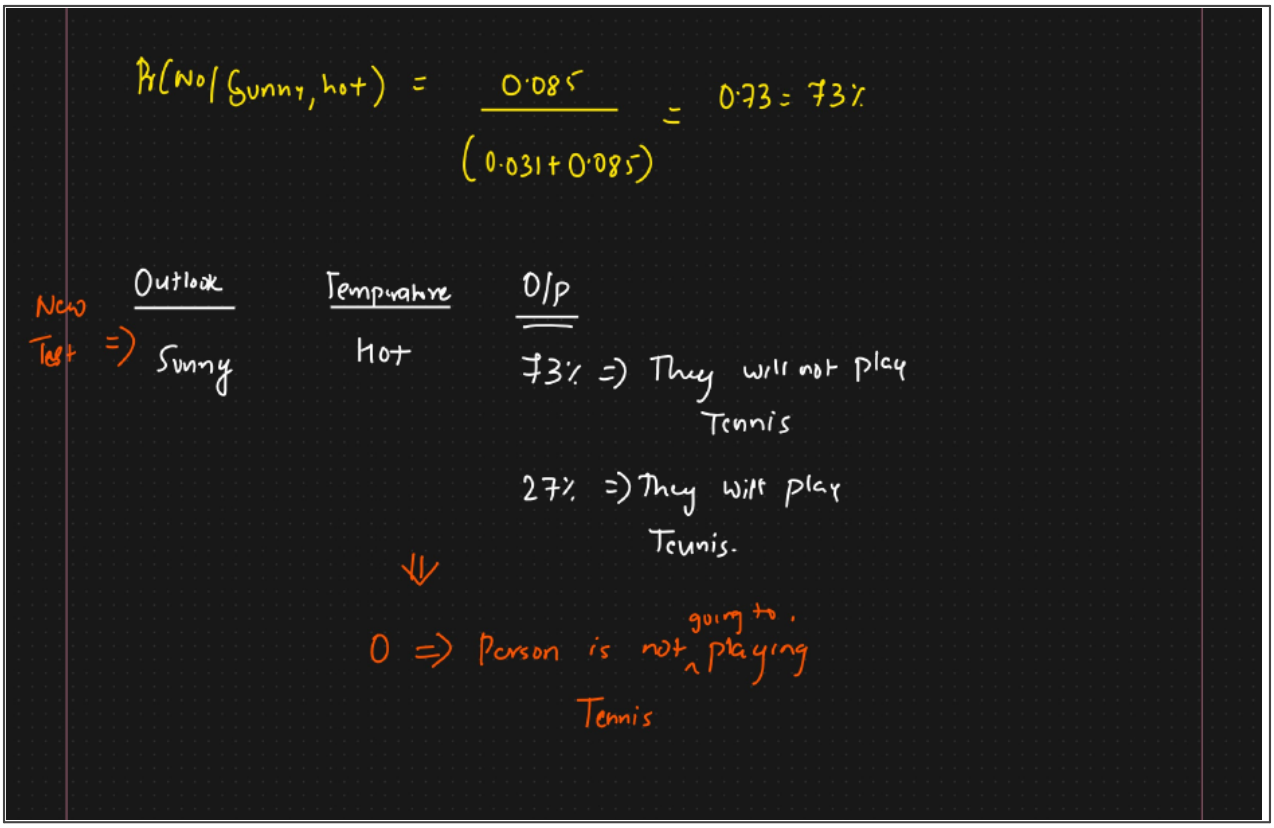
	Yes	No	P(E/Yes)	Pr(E/No)
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rain	3	2	3/9	2/5

	Yes	No	P(E/Yes)	P(E/No)
Hot	2	2	2/9	2/5
Mild	4	2	4/9	4/5
Cool	3	1	3/9	3/5

		P(Yes)	P(No)
Yes	9	9/14	5/14
No	5		

Isme dikhaya gaya hai: Play Tennis dataset, Outlook aur Temperature ki frequency tables, P(Yes) = 9/14 aur P(No) = 5/14, aur (Sunny, Hot) ke liye Yes ka score (2/63 = 0.031) tatha No ka score (3/35 = 0.085).

Original Handwritten Notes — Naive Bayes, Page 5



Isme dikhaya gaya hai: Normalization se P(No) = 73%, aur final conclusion ki new test (Sunny, Hot) ke liye person tennis nahi khelega.

6. Bonus — Zero Probability Problem aur Laplace Smoothing

(Ye section teacher ke notes me nahi hai, lekin exam aur practical dono ke liye bahut important hai.)

Dhyaan se dekho Outlook table: **Overcast** ke liye No me count **0** hai, yaani P(Overcast | No) = 0/5 = 0. Ab agar kabhi test data me Overcast aaya, to No ki poori multiplication me ek term 0 hone se **poora score 0** ho jaayega, chahe baaki features No ke kitne bhi favour me hon. Ye achha nahi hai — sirf ek unseen combination se class poori tarah reject ho jaati hai.

Solution — Laplace (Additive) Smoothing: har count me **1 add** kar do, aur denominator me categories ki sankhya (k) add kar do:

> **P(value | class) = (count + 1) / (class ka total count + k)**

Example: Outlook me 3 categories (k = 3) hain, to P(Overcast | No) = (0 + 1) / (5 + 3) = **1/8**, ab 0 nahi hai. Scikit-learn me ye **alpha** parameter se control hota hai (default alpha = 1).

Part 2 — Variants of Naive Bayes

Feature ke type (binary, text ya continuous) ke hisaab se Naive Bayes ke teen variants hote hain.

7. Variants Ki Zaroorat Kyun?

Section 5 me hamare saare features **categorical** the (Sunny, Hot...), isliye frequency table se probability nikaal li. Lekin real data hamesha aisa nahi hota — kabhi features sirf 0/1 hote hain, kabhi text hota hai, kabhi height/weight jaise continuous numbers. Isliye **P(feature | class)** nikalne ka tarika feature ke distribution ke hisaab se badalta hai. Teacher ne teen variants bataaye hain:

#	Variant	Kab use karein	Feature kaisa hota hai
1	Bernoulli Naive Bayes	Features Bernoulli distribution follow karein	Sirf 2 values: 0 / 1
2	Multinomial Naive Bayes	Input text ho	Word counts / frequencies
3	Gaussian Naive Bayes	Features Gaussian (normal) distribution follow karein	Continuous numbers

8. Bernoulli Naive Bayes

Jab bhi tumhare **features Bernoulli Distribution** follow karte hon, tab Bernoulli Naive Bayes use hota hai. **Bernoulli distribution** matlab sirf do hi possible outcomes — **0 ya 1** (jaise haan/na, pass/fail, aaya/nahi aaya).

Teacher ka dataset: teeno features ke sirf do-do hi values hain, aur output bhi Yes/No hai:

f1	f2	f3	Output
Yes	Pass	Male	Yes
Yes	Fail	Female	No
No	Pass	Male	Yes
Yes	Fail	Female	No

Computer ke liye isko 0/1 me convert kar dete hain (mapping koi bhi chalegi, bas consistent ho): **Yes = 1, No = 0, Pass = 1, Fail = 0, Male = 1, Female = 0**.

f1	f2	f3	Output
1	1	1	Yes
1	0	0	No
0	1	1	Yes
1	0	0	No

> **Yaad rakhein:** Bernoulli NB me har feature **binary** hota hai. Probability nikalne ka tarika wahi hai jo section 5 me tha — bas har feature ki sirf do values (0 aur 1) ke liye P(feature | class) count se nikalte hain.

9. Multinomial Naive Bayes

Jab **input text** ho (jaise email, review, message), tab **Multinomial Naive Bayes** use hota hai. Teacher ka example hai **Spam Classification** — input feature **Email Body** hai aur output **Spam / Not Spam (Ham)**:

Input (Email Body)	Output
You have Million $$ lottery	Spam
KRISH you have done good job	HAM (Not Spam)

9.1 Text Ko Numbers Me Kaise Badlein?

Machine learning model **text nahi samajhta**, sirf numbers samajhta hai. Isliye pehle text ko **numerical values (vectors)** me convert karna padta hai. Ye **Natural Language Processing (NLP)** ka kaam hai, aur teacher ne teen techniques likhi hain:

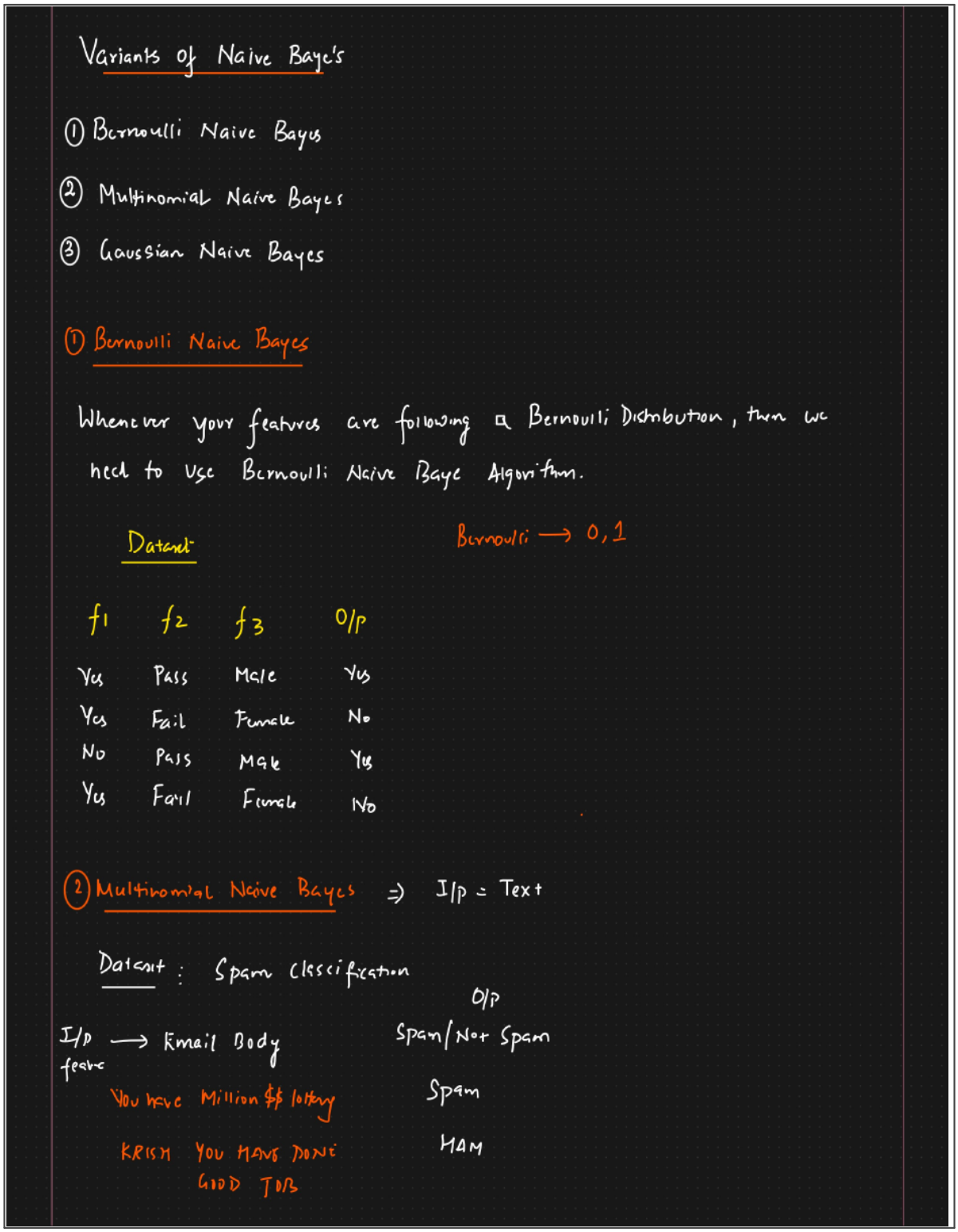
Technique	Kaise kaam karta hai
1. BoW (Bag of Words)	Poori vocabulary banate hain aur har sentence me har word kitni baar aaya, wo count karte hain. Word ka order ignore hota hai.
2. TF-IDF	BoW jaisa hi, lekin common words (jaise "you", "have") ko kam weight aur rare/important words ko zyada weight milta hai.
3. Word2Vec	Har word ke liye ek dense vector banta hai jo word ka meaning capture karta hai (similar words ke vectors paas hote hain).

Chhota BoW example: vocabulary = [you, have, million, lottery, krish, done, good, job]

Sentence	you	have	million	lottery	krish	done	good	job
You have million lottery	1	1	1	1	0	0	0	0
Krish you have done good job	1	1	0	0	1	1	1	1

Dhyaan dein: Multinomial NB word **counts/frequencies** par kaam karta hai, isliye ye BoW aur TF-IDF ke saath natural fit hai. Word2Vec ke vectors me negative values bhi hoti hain, jo scikit-learn ka MultinomialNB accept nahi karta (wahan aam taur par Gaussian NB ya koi aur model use hota hai).

Original Handwritten Notes — Variants, Page 1



Isme dikhaya gaya hai: Naive Bayes ke 3 variants (Bernoulli, Multinomial, Gaussian), Bernoulli ka dataset (f1, f2, f3 → O/P), aur Multinomial NB ka spam classification example (Email Body → Spam / Ham).

10. Gaussian Naive Bayes

Jab **features continuous numbers** hon aur **Gaussian (Normal) Distribution** follow karte hon, tab **Gaussian Naive Bayes** use hota hai. Gaussian distribution wo **bell-shaped curve** hai jiska peak mean (μ) par hota hai. Teacher ka dataset: Age, Height, Weight jaise continuous columns aur output Yes/No. Iris dataset iska classic example hai (petal/sepal ki lengths continuous hoti hain).

10.1 Kaise Kaam Karta Hai?

Continuous values ki frequency table nahi ban sakti (kyunki har value alag ho sakti hai). Isliye har class ke liye aur har feature ke liye training data se **mean (μ)** aur **standard deviation (σ)** nikaalte hain. Phir naye value x ke liye Gaussian formula se **P(x | class)** nikalte hain:

$$P(x \mid \text{class}) = \frac{1}{\sqrt{2\pi\sigma^2}} \times \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Baaki process bilkul same hai: sabhi features ki values ko class ke prior ke saath **multiply** karo, dono (ya saari) classes ke scores compare karo, aur jiska score bada ho wo prediction.

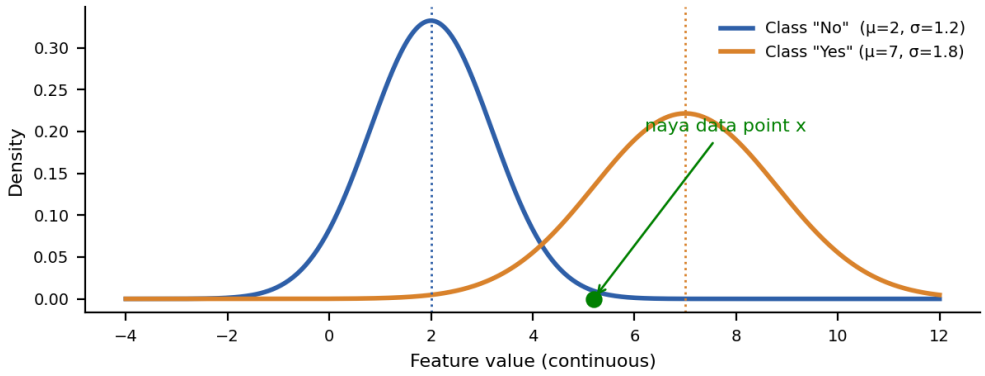
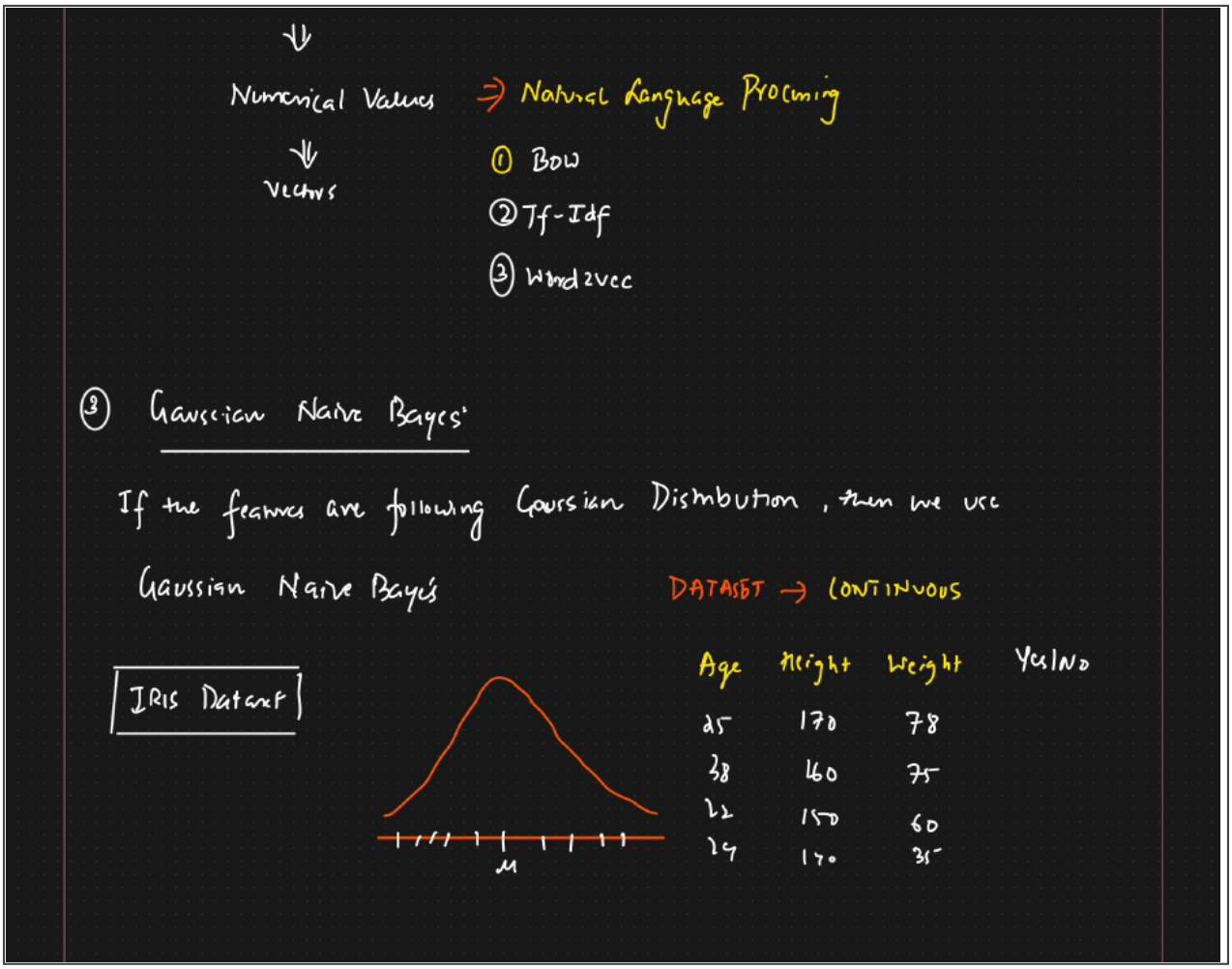


Figure: Har class ka apna bell curve (apna μ aur σ) hota hai. Naya data point jis class ke curve par zyada "ooncha" baithta hai, us class ka P(x | class) bada hota hai.

> **Dhyaan dein:** Continuous feature me P(x | class) technically **probability density** hoti hai, exact probability nahi. Lekin classes ko compare karne ke liye wahi kaafi hai, isliye algorithm bilkul theek kaam karta hai.

Original Handwritten Notes — Variants, Page 2



Isme dikhaya gaya hai: Multinomial NB me text ko vectors me badalna (BoW, TF-IDF, Word2Vec) aur Gaussian Naive Bayes — continuous dataset (Age, Height, Weight), bell curve aur Iris dataset ka reference.

11. Teeno Variants Ki Comparison

Point	Bernoulli NB	Multinomial NB	Gaussian NB
Feature type	Binary (0 / 1)	Counts / frequencies	Continuous values
Distribution	Bernoulli	Multinomial	Gaussian (Normal)
Teacher ka example	Yes/No, Pass/Fail, Male/Female	Spam classification (email text)	Iris; Age, Height, Weight
Input ko kaise taiyaar karein	0/1 encoding	BoW / TF-IDF se text → vector	Seedha numbers (mean, std nikalte hain)
Scikit-learn class	BernoulliNB	MultinomialNB	GaussianNB

12. Kab Konsa Variant Chunein?

- Features **sirf 0/1 (haan/na)** hain → **Bernoulli NB**
- Input **text** hai ya word **counts** hain → **Multinomial NB**
- Features **continuous numbers** hain (height, weight, iris measurements) → **Gaussian NB**
- Zero probability ka darr ho (kisi value ka count 0) → **Laplace smoothing** (alpha) lagao

13. Poore Notes Ka Quick Revision

Topic	Yaad rakhne wali baat
Independent events	P(A and B) = P(A) × P(B)
Dependent events	P(A and B) = P(A) × P(B | A)
Bayes' Theorem	P(A | B) = P(A) × P(B | A) / P(B)
Naive assumption	Class diya ho to saare features independent
Naive Bayes formula	Score(y) = P(y) × P(x_1 | y) × P(x_2 | y) × ... (denominator constant, cut kar do)
Prediction	Jis class ka score sabse bada, wahi answer (normalize karke % bhi nikaal sakte hain)
Play Tennis (Sunny, Hot)	Yes = 27%, No = 73% → **No**
Zero probability	Laplace smoothing: (count + 1) / (total + k)

Ek line me yaad rakhna: Naive Bayes = **Bayes' Theorem + "features independent hain" ka assumption**. Har class ka score nikalo (prior × conditional probabilities), jiska bada ho wahi prediction. Feature type ke hisaab se Bernoulli (0/1), Multinomial (text) ya Gaussian (continuous) chuno.